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On the island of Sulawesi in Indonesia, humans painted pigs on the walls of their cave dwellings some 51,200 years ago. Since that time, creativity and artwork has primarily been the domain of our species. With the development of generative artificial intelligence; music, visual art, writing, and a myriad of other acts of creativity can now be done with a few prompts from any neonate. Humanity stands now at the dawn of a new age in the creative world. Machine can create what once only man alone could make. Lawmakers face the terrible burden of being reactionary agents in times of change. Until a problem has arisen to the appropriate level in the cultural zeitgeist, laws tend to fail to address the intricacy of any situation because lawmakers simply don't know the future. The need to catch up is immediate because the technology continues to develop, advance, and grow. The courts have decided on or been presented debate on a number of cases in front of them which use logic that I will discuss in this paper. However, this lacks the full endorsement and scrutiny of public stakeholders, instead relying solely on the bench's discretion. An appropriate legal framework for how to address the traditional stance of intellectual property rights both in the training and in the generated product of this burgeoning technology has not yet been passed by any lawmakers.

Thaler v. Vidal created the first legal decision in determining the copyright of arts created using artificial intelligence. Circuit Judge Stark wrote for the majority in deciding that the U.S. Patent Act applies only to the works and creations of a human actor, but failed to set a test as to the proportion of work a human must take for an artificial intelligence generated copyright to become valid under the Patent Act. Mr. Thaler initiated another important decision in Thaler v. Perlmutter. The Court further clarified that the Copyright Act of 1976 requires the creator to be human and rejects the application of works for hire status to an artificial intelligence's copyright being transferable to the human that prompted it.

The preclusion that artificial intelligence is ineligible to copyright its own work and thus unable to transfer those rights to any human actor is a short bridge in logic. Judge Howell specifically notes that, "By its plain text, the 1976 Act thus requires a copyrightable work to have an originator with the capacity for intellectual, creative, or artistic labor. Must that originator be a human being to claim copyright protection? The answer is yes.", and in the paragraph following further clarifies that because of the lack of capacity in artificial intelligence, it would find no incentives and have need for the protection of copyright.

Humans and other animals had to first go through eons of evolution and small steps to develop symbolic cognition. Artificial intelligence gets to skip some steps in this capacity though. Humans are able to take portions of the skill that we possess to utilize symbolism and train an artificial intelligence to perform similar creative tasks, with the fault that the artificial intelligence is not able to recognize the symbolic reasoning involved in human artistic expression yet. Due to Mr. Thaler's filings, the Court further discusses artificial intelligence as a tool, but does not establish a clear test as to the degree of input required from a human at generation or afterwards to be eligible for copyright.

Currently artificial intelligence requires examples in order to learn how to create art in its various forms. In Andersen v. Stability AI Ltd and RIAA v. Suno, artists and other representatives from creative industries are asking the courts to decide on the rights of those who had their work used for the training of artificial intelligence systems.

Legislators from the United Nations and the United States are just beginning to delve into the question of the rights of creators and trainers and of the production of art made by artificial intelligence with frameworks and resolutions. In October of 2022, Senator Thom Tillis of North Carolina and Senator Chris Coons of Delaware requested the U.S. Patent and Trademark Office and the U.S. Copyright Office to assemble a commission to make determinations as to the future of artificial intelligence and the respective intellectual properties their offices cover. In a similar vein in May of 2023, Representatives Scott Fitzgerald of Wisconsin, Darrell Issa of California, Ben Cline of Virginia, and Nathaniel Moran of Texas in their capacity as members of the House Judiciary Subcommittee on Courts, Intellectual Property, and the Internet wrote to Register Perlmutter of the

U.S. Copyright Office seeking clarity on the position and involvement of the Office in guidance and authority to address future issues of copyright and artificial intelligence.

Letters of response were sent to Senators Tillis and Coons in December of 2022, and to the members of the House Judiciary Subcommittee on Courts, Intellectual Property, and the Internet in May of 2023. In these letters, the U.S. Patent and Trademark Office and the U.S. Copyright Office clarified that while they were not forming the requested commission, they are involved in the National Security Commission on Artificial Intelligence and are actively engaging stakeholders across the industry, public, and other governmental organizations in discussion and decisions that would provide guidance these matters. Register Perlmutter reiterates this sentiment in her letter to the House Judiciary Subcommittee on Courts, Intellectual Property, and the Internet. A concerning matter arises at the end of her letter though, unless specifically empowered by legislation, a regulatory agency such as the U.S. Copyright Office is not given powers to act as law enforcement or as the courts in pursuing violations that exist outside of their statutory authority.

This paper is an examination of standing laws, case precedence, and models of intellectual property law as it relates to generative artificial intelligence. By addressing the concerns of companies, consumers, and programmers, I will lay out historical standings that are applicable to artificial intelligence and propose rights for the stakeholders interacting with this burgeoning technology. My goal is to propose frameworks for legislators that would outline how much human involvement is required and give adequate protections to consumers and creators, while maintaining protections to continue encouraging further development and deployment in the public sphere.

It is important to balance the ability of a company or anyone using artificial intelligence as an end user to retain some level of copyright-esque protection for the outcomes of their ideas. However, I would also caution legislators in favor of more rights to those who have had to interact with a medium and created their works in the “old fashioned” way as the law has historically held. Full protection of works generated by artificial intelligence could create an imbalance between those who had to develop skill and create in a physical medium against those who, once the artificial intelligence is trained, can merely enter a prompt and generate more works than a human artist could in a lifetime.

My research is rooted in finding an equitable balance between the interests of creators and generators. I am examining the ways through which copyright is normally viable, i.e., creating works through physical mediums and non-generative artificial intelligence digital creation or alteration, and seeking the ways to either adapt or create new pathways to viability.

While the D.C. Circuit has offered a sentiment on works for hire and the idea that artificial intelligence couldn’t be treated like an employee granting copyright to its employer, I believe this may not be an entirely unviable avenue of exploration. It does open questions to timelines though. After all, what marks the birth or death of a program? Lifetime plus 70 years could be a very long time and stifling to competition and human ability. However, failing to grant any path to copyright would dissuade investment and development of these technologies as they would have little marketability for the works they create.

Further, the question of compensation for works used in the training of artificial intelligence still needs to be examined. While the courts are opening books on cases which require it to be asked, I am also taking the question of fair use laws in copyright to assist in determining if training artificial intelligence should be considered any different than training human artists. Exceptions exist within the Digital Millenium Copyright Act, the Lanham Act, Patent Act, and Copyright Act, for the fair use training of humans in all of the intellectual properties covered by said acts.

Generally, the creators whose work is used in the training of humans have received compensation in some other form: fame, accreditation, awards, or financial compensations. While the training of the computer may not warrant complete exception, depending on methodology there may be other venues that could be appropriate and continue to encourage development of the technology without hampering the marketability of the compensated creators or any stakeholders right to contract.

There currently stands no statutory law that directly answers the question of copyright in the matters of artificial intelligence’s inputs and productions. This is leading to issues within the market. From companies using the work of human artists without compensating them for the time and effort that went into their original productions, to companies telling end users that the creations coming out of their prompts to artificial

intelligence engines are fully copyrighted and protected in the generator's ownership. None of the current case law has made it high enough in the court system to create a unified stance on either portion of this question. Without a solid answer, artists and generators are for lack of better terminology being robbed of the benefits that should be available to them under any other circumstance of the arts creation and usage.

My research presents a balanced recommendations that legislators could fit into the next generation of artificial intelligence legislation to cure these ills. Currently passed laws have already addressed in legislation the right to data privacy and protection, to the copyright of the coding and artificial intelligence itself, and even touch upon development of new protocols and products to include in future generations of artificial intelligence. However, by failing to address inputs and products of artificial intelligence there stands a gaping hole in protections and discussion. Filling this gap gives the opportunity for endorsement from the artistic community in the future development of generative artificial intelligences, and answers with certainty that they are protected and able to continue working without fear of losing their place in the marketplace both in terms of their unique skills and styles and in terms of their proper financial compensation. It would also be a more accurate representation of the voice of the people for the choice of protections in generated artworks to be coming from Congress than from decisions of the Courts.

This will open a new field of artistic expression for those who previously lacked the skill with a brush or instrument to participate in the artistic community. It further allows for new developments such as the artificial intelligence based musical artist Noonoori that was signed to a record contract with Warner Brother's in 2023 to have certainty in their investment that they have met an appropriate guideline of human input in order to have a copyrightable product.

While I find the efforts and work of the U.S. Copyright Office and other governmental and private stakeholders in this matter to be laudable and in good faith, the simple truth is that their guidance falls short in enforcement powers without Congressional backing, and in light of *Loper Bright Enterprises v. Raimondo* a future court could find reason to disagree with these organizations' guidelines. Further, while the U.S. Copyright Office now has the Copyright Claims Board, it is both voluntary and has provisions for trial in Federal court instead of and after a decision is issued by the Board. Making the Copyright Claims Board and the guidance of the U.S. Copyright Office useful and compelling in regard to artificial intelligence and copyright status, but still lacking the authority that the decisions of the courts or passage by lawmakers into statute would hold.

At several points in this paper, I have and will again refer to synthetic intelligence. I use this term to describe an artificial intelligence that has advanced far enough to pass beyond programmed and imitation intelligence and has gained awareness and sentience of itself. It is easier to point to science fiction for examples of artificial intelligence that has advanced into sentience in the way that humans recognize each other as sentient beings. From Asimov's Three Laws of Robotics and Wall-E in the positive to neutral Androids of Phillip K. Dick to the destructive Matrix and Skynet, science fiction has long given the world images of machines that are as smart and capable of understanding the world and their place in it as their human creators. While we may be nearing the end of the viability of quoting Moore's Law, technology advances as a breakneck pace and our understanding and ability to create new technologies continuously moves forward and builds upon itself. While the definition of what a synthetic intelligence will look like is beyond what I am able to delve into philosophically and technologically without derailing this paper, it is a real possibility that society may see the first of this new form of intelligence in the lifetime of children born today. Discussing and determining what protections human intellectual labors deserve in relation to a new technology that can outpace and replace said labors and how we define the boundaries of those protections will be important in understanding how to approach the question of rights for the intellectual labors of a synthetic intelligence when it arrives.

My primary concern is to the flesh and blood human artists though. They need this issue addressed in order to fully enjoy the protections they are due for their intellectual labors. But, I'd be remiss to fail in keeping an eye on the horizon for issues we may not have addressed properly yet.

The Lanham Act, Pub. L. 79-489.

The Lanham Act is a staple of the American intellectual property law, representing the rules and guidelines for trademarks. While it lacks persuasiveness towards the actual discussion of artificial intelligence and art in this paper, it is of great value in understanding the structuring and writing in American intellectual property law, and thus grants insight upon writing a model act or amendment to standing acts on the matter.

The Copyright Act of 1976, Pub. L. 94-553.

I believe the U.S. Copyright Act of 1976 to be the persuasive law on the books for the matter. While I am dealing with artificial intelligence, the works being used as input and being generated as output are works of more traditional medias such as images and music which are covered by the Copyright Act. In the case of inputs, it provides guidance upon what inputs programmers can use in training and provides guidelines to preventing infringement during the creation of new artificial intelligence engines. In the case of outputs, it provides a goal post for generators to reach for as they attempt to determine what would be necessary for them to find copyright protection on works that began as prompts into a generative artificial intelligence.

The Digital Millennium Copyright Act, Pub. L. 105-304.

Initially, the Digital Millennium Copyright Act may appear to be the governing ruleset for training and the output of artificial intelligence. However, I believe the Digital Millennium Copyright Act focuses upon the programming and coding of the artificial intelligence but fails to address traditional media copyrights in terms of being created by artificial intelligence. Depending upon the methodology of the training, a Digital Millennium Copyright Act violation claim could be appropriate, but may fall upon the same issues as *Viacom International, Inc. v. YouTube* in which a large-scale operation such as YouTube could make the claim that while they attempt to enforce existing copyright rules, they cannot always guarantee that the users of their database do not manage to circumvent their protective measures and post copywritten works onto the database. Such usage should also be considered in terms of how an artist can study the works of others via platforms such as Pinterest or DeviantArt and learn to create similarly styled works of their own.

The U.S. Patent Act, 35 U.S.C.S. § 1 et seq.

The U.S. Patent Act is the final primary body of American intellectual property law that I am using for examining structure and application. It presents useful understanding on the structure of our intellectual property laws and was a factor in another persuasive decision on the matter of artificial intelligence and the works it can create. Depending upon the method of programming for an artificial intelligence engine though, I believe this act may be persuasive for dealing with inputs taken from human artists.

The Copyright Alternative in Small-Claims Enforcement Act of 2020, Pub. L. No. 116–260, Div. Q, Title II, § 212.

Beginning on page 997 of the Consolidated Appropriates Act of 2021, one finds the Copyright Alternative in Small-Claims Enforcement Act of 2020. The primary importance of this act to my research is understanding the establishment of the U.S. Copyright Office's Copyright Claims Board and the extent of enforcement power that was granted to this subsection of the Office. The Board was created as a voluntary alternative to relieve some of the burden upon Federal courts, as all copyright claims must go through Federal courts. It maintains a 3-year statute of limitations in alignment with the U.S. Copyright Act, and generally has comparable powers to a Federal Court in conducting findings of fact and making determinations within their purview. However, the Copyright Claims Board does not preclude action in the Federal courts, and thus while it provides solid guidance, does not prevent any action taken within the court being appealed up to standard Courts. This particular addition shows a strong attempt at dealing with copyright claims and trying to alleviate the wait and ponderance in making decisions, but is ultimately leaves the board lacking in authority when compared to Federal Courts or Congress.

*Andersen v. Stability AI Ltd.*, 700 F. Supp. 3d 853 (2023).

Representing the initial case for copyright infringement in the training of artificial intelligence, this case is a crucial starting point in understanding, in legal terms, the complaint of human artists in having their works used for the training of art generating artificial intelligence. It addresses the complaint that the artwork is being stored and compiled for the artificial intelligence to use in creating such works. A primary question that arises from this case is if the artwork is compiled and stored as digital copies of the copyrighted works, or if the work is studied, converted into an algorithmic format, and then the algorithm is used by the artificial intelligence. The first issue would represent clear violation of copyrighted materials, but the later would require a new understanding on the question. If a work is converted into algorithm, is it in essence still the copyrighted picture, or is it more akin to how a human can remember seeing a Van Gogh and attempt to replicate his painting style with their own hands.

*Andy Warhol Found. for the Visual Arts, Inc. v. Goldsmith*, 598 U.S. 508 (U.S. 2023).

While I may agree more with the dissent in this case, this represents the prevailing decision on the matter of transformative art. The decision focuses primarily upon that parody is a valid reason for copying but that satire must make a meaningful distinction from the original work and that the works cannot be in competition with one another for purpose such as the commercial space. The standard of review that is shown by the court in this instance is a rather high bar, both when looking at the purposes through a limited scope such as commercial meaning any attempt for money rather than looking to the distinct markets that would place different values upon the works and when performing an examination of the differences in artistic choices within a piece such as tone and composition. As the binding case law on the matter though, it gives a deeper understanding as to the definition of purpose for fair use and in examining how to address the level of transformation necessary for artwork to utilize pre-existing works and to draw comparison to the inputs and outputs of generative artificial intelligence.

*Andy Warhol Found. for the Visual Arts, Inc. v. Goldsmith*, 598 U.S. 508 (U.S. 2023). (Kagan, dissenting).

Justice Kagan's argument against the precedent being set by the majority is not the binding precedence being set by this case. However, she provides an excellent dissection of issues for further examination of transformative art. Kagan sets out a number of comparisons that would reasonably seem to also fail the test of transformative set by the majority that represent music, visual arts, and writing. Comparing the majority and dissenting opinions highlights to me the need for clarification by Congress and the difficulty of finding balance when addressing issues of opinion such as artistic tastes.

*Bleistein v. Donaldson Lithographing Co.*, 188 U.S. 239 (U.S. 1903).

This case was an examination of copyright law in the commercial space and set the precedence that commercially created works are worthy of copyright protection. It is the source of a famous Oliver Wendell Holmes Jr. quote related to the nature of the judicial branch and its responsibility to examine only matters of the law and to refrain from attempting to also be art critics. In settling the matter of copyrights for a pair of posters created for a circus, it was established that there was an enforceable copyright that belonged to the lithographic company for said posters. It established that the rights for copyright were independent of the customers idea of the posters even where the images could be based on vague designs there is still a uniqueness in the creation of said content that warrants protection in favor of the creator of the work as opposed to the idea. The case provided a heritage and provided clarity in understanding how delicate of a balance it can be for Courts to simultaneously perform their duties as interpreters of law while also maintaining the public interest in matters that require specialized skills or are matters of taste.

*Burrow-Giles Lithographic Co. v. Sarony*, 111 U.S. 53 (U.S. 1884).

This particular case is roughly a century and a half old. I found the case enlightening despite this age though. Reading through the opinion of Justice Samuel Freeman Miller, I found myself able to see a clear picture being made of a balanced opinion on how to handle a technology that had only come into prominence and usage amongst the public within the lifetime of the parties involved in the case. Miller lays out that the act of choosing what Wilde wore, how he was posed, what angle to take the picture, and every decision that occurred between the decision to take the photograph and when Mr. Sarony triggered the device to take said photograph. It gave me a clearer picture of how many decisions in any creative process could be a quantifier in discussing the uniqueness of a protected piece. It is important to these questions in that photography was a young technology that allowed the creation of a copyright protected piece of work at an expedited pace with less physical and temporal exertion on the artist than had previously been required of artists.

*Doe v. Github, Inc.*, 672 F. Supp. 3d 837 (N.D. Cal. 2023).

There was an order issued by Judge Tigar of the U.S. District Court for the Northern District of California on September 27, 2024 showing there may be an appeal to the 9th Circuit coming for this case. During the dismissals in this case, it was shown that there may be a need for copyright claims of infringement to include actual reproductions of copyrighted materials in order to prove that infringement occurred during the training process that I will discuss more fully later on in reference to ongoing cases such as Andersen and Tremblay. A major difference between this case and any other that I examined in relation to inputs for training of artificial intelligence is that Doe involves coding across the board, rather than a question of if art turned into code qualifies for copyright. It gives a clearer understanding of the requirements the Courts are looking for without the muddying factor of conversion.

Eldred v. Ashcroft, 537 U.S. 186 (U.S. 2003).

Justice Ruth Bader Ginsburg focuses on issuance of previous limits and changes to the limits by other acts of Congress to justify the Constitutionality of extending copyright terms via the Copyright Term Extension Act. The first set of reasoning this case covers is that due to an increase in how long people are living gives good reason to also increase how long copyright protections last. A second set of reasoning in this case examines the question of definition and determines that as long as a finite number is set Congress has performed its duty in setting a limit to the term of copyright. This provides clarity in relation to artificial intelligence as it grants Congress can set definitions and change them appropriately to mitigate new challenges as they arise and helps focus thinking as to the appropriateness of time when looking at human life expectancy and at how that is a very different question for the life expectancy and a non-living entity such as artificial intelligence.

Feist Publ'ns, Inc. v. Rural Tel. Serv. Co., 499 U.S. 340 (U.S. 1991).

The Court was faced with the question of facts in copyright and fair use. It clarified that the notion that just because someone put in time and effort into performing a task that could be copyrightable does not mean that the work itself is worthy of copyright. The establishment that raw data cannot be copyrighted in and of itself provides crucial insight when looking at inputs for artificial intelligence. The usage of data may not rise to copyright infringement based upon this case, but there are other factors such as the arrangement that needs to be examined when looking at the issue. It further assisted in granting insight to how the Courts examination the advancement of culture and science intent of copyright.

Kelley v. Chicago Park Dist., 635 F.3d 290 (7th Cir. 2011).

Kelley takes a hard look at copyright when dealing with aspects outside of the artists control and if that unpredictability can be a disqualifying feature to copyright. The art at question was a garden specially designed and curated by the artist for the City of Chicago. This case makes limitations as to installation and site-based works of art and what quantifies as a visual work for the purposes of copyright protections. A portion of the ruling makes special highlights that as plants are living organisms that grow and change in ways beyond the artists control, copyright cannot give protection because the changes do not reflect the original vision of the artist or the work at the time of installation. It grants insight that chaotic elements beyond an artist's control do have weight in determining qualification for protection. However, I find the notation of changes occurring after the work is created to be a differentiating factor when compared to generative artificial intelligence and the unpredictability of what the algorithm will produce to any prompt given. I will address this further during my legal analysis, as there are a number of examples to be made in which currently protected artforms utilize chaotic and unpredictable elements in the process of initial creation of copyrightable works.

Loper Bright Enters. v. Raimondo, 144 S. Ct. 2244 (U.S. 2024).

A very recent landmark decision that overturned Chevron v. NRDC. This case had primary relevance to my research in establishing the limits of the agencies governing intellectual property law to make meaningful determinations in relation to their covered bodies of law. On one hand, the decision unbinds the Courts to hear additional input from experts outside of governmental agencies when making decisions in the cases before them. However, it also takes some authority away from these agencies which operate daily in their relevant fields as experts and at the direction of Congress and the executive branch. The freedom of the Court to operate independently and to perform full and thorough examinations is of utmost importance, but only time will tell if this hurts the ability of the agencies to issue guidance effectively to the various industries they effect.

Obodai v. Demand Media, Inc., 2012 U.S. Dist. LEXIS 83109 (S.D.N.Y. 2012).

The first application of stricter questions asked by Viacom in order for databases and social media platforms to protect themselves from copyright infringement liability. It stands in stark contrast though, as it shows that a company can easily implement a guideline policy that not only keeps them free of liability, but also actively functions both in terms of the technology operating as desired and in encouraging users to be actively aware of their actions and the consequences. In studying this case, I particularly focused on the definitions being set by the court's opinion.

SocialMediaOne, Noonouri: virtual influencer, digital model and international Instagram star, <https://socialmediaagency.one/noonouri-virtual-influencer-digital-model-and-international-instagram-star/> (last visited 8/4/2024)

While not a scholarly work, this source has assisted as a matter of perspective. Noonooori is a possible future of artificial intelligence in the creative space. The character is a conglomeration of her creators' vision; software such as artificial intelligence to generate lyrics and replicate the full or partial body of the music in terms of vocals and instrumentals; computer-generated imagery and artificial intelligence to create videos and other visual representations of the "artist"; and the hard work of the creator and marketing team to create a seamless vision of essentially an artificial intelligence musician and media personality. *Thaler v. Perlmutter*, 687 F. Supp. 3d 140 (D.D.C. 2023).

The current standing case on the subject of artificial intelligence and Copyright and representing the only current District ruling on the matter. In this case, Thaler attempted to copyright images created by his "Creativity Machine" and was denied on the basis that copyright law currently requires a human creator. It shows that case law has given a history to the human requirement going back to 1884 with *Burrow-Giles Lithographic Co. v. Sarony*. Burrow-Giles established a photographer held rights to works created by the taking of a photograph, and the human element required in that act despite the use of a camera. The court left the same question as in *Thaler v. Vidal* of how much human interaction is required to make a work copyrightable and has left such determination to a future court or to lawmakers to provide clarity upon. *Thaler v. Vidal*, 43 F.4th 1207, 1213 (Fed. Cir. 2022).

A similar attempt by Thaler to claim intellectual property rights on an artificial intelligence creation that would generate plans for patentable products. It met the same response to *Perlmutter* and *Thaler v. Hirshfeld* that a human is required, and was left with a similar question of the percentage of human interaction is required. It gives a depth of perception upon the rulings of another bench at a similar level to the *Perlmutter* argument, showing that the pervasive stance of the courts is likely to run along this line should such questions be risen before the Supreme Court.

*The Recording Industry Association of America v. Suno, Inc.*, 2024 U.S.D.Mass. (Filed June 2024).

This case presents the music industry attempting to protect its creative rights and to establish their claims within the realm of art generated via artificial intelligence. As persuasive and binding as the rulings from this case will be upon final ruling, I believe the case is most powerfully used in understanding what companies and industry stakeholders as a whole are seeking from artificial intelligence. In their claims, the Recording Industry Association of America discusses works already in progress within their respective spheres to integrate and encourage the development of artificial intelligence within the music industry. I seek to find a balance between providing absolute protection of the individual artist, and also providing a measure of protection and incentive to the industry to continue investment and integration. Understanding the views of large companies within this sphere such as Sony, Capitol Records, and Warner Music provides a great deal of insight as to what they would seek from any new law dealing with creative works and artificial intelligence.

*Tremblay v. OpenAI, Inc.*, 2024 U.S. Dist. LEXIS 141362 (N.D. Cal. 2024).

*Tremblay* represents the second major case that is currently ongoing which deals with artificial intelligence being trained with works that may have grounds for copyright infringement claims. In examining the filings of *Tremblay* in comparison with *Andersen*, I was able to find a clearer picture as to the hurdles that are facing the creative community in dealing with artificial intelligence inputs. It highlights that the Courts may not have the ability to address concerns on this matter as statutory law has not provided adequate protections, methods for discovery, or even definition of terms and limits for generative artificial intelligence training. *Urantia Found. v. Maaherra*, 114 F.3d 955 (9th Cir. 1997).

In reviewing cases and articles related to artificial intelligence and copyright, this particular case seems to manage to be referenced by almost every Judge and author. I am unsure on how to best handle my writings on the matter. The case is a fairly common claim within the religious and spiritual spheres, a book was written and the claim of the faithful is that the work was supernaturally inspired and that humans merely heard and wrote out what they were told. Judge Mary Schroeder is extremely sensitive to the beliefs of the parties and takes a measured approach to attempting to apply a standard review of the copyright claim.

*Viacom International, Inc. v. YouTube, Inc.*, 676 F.3d 19 (2nd Cir., 2012).

*Viacom V. YouTube* outlined early metrics for future use as to what a properly monitored social media platform would need in order to meet reasonable defensibility in a claim of copyright violations. Viacom sued YouTube on the grounds that the then 5-year-old company had billions of views daily and that users were

illegally uploading their protected materials to the platform. Initially, YouTube did seem to have some reasonable deniability. They did not actively direct the traffic on the site, instead using a simple algorithm that made recommendations for new videos based on the tags of your previously watched videos, and did not give active instruction or encouragement to upload videos that were in violation of copyright. However, it was uncovered that there were employees at the company who had knowledge of these violations and had communicated about it in their capacity as employees of YouTube but had failed to take any action to attempt to remove and discourage future violations by the violating users. There is naturally a discussion to be had about corporate ethics and employee engagement in relation to this case as well. However, this case forced a conversation as to what a company should do as database or social media providers of this type of material. As this paper asks two separate but intertwined questions, I will ask the questions of rule in two parts. The first and easier to tackle is: are the works produced by artificial intelligence able to receive copyright protections under current law? The answer is simply no. The statutory law of the matter is entirely silent. Congress hasn't passed any laws that directly give the ability to artificial intelligence, and our laws have the justifiable history of being solely focused on a human creator for art. *Thaler v. Perlmutter* is the only case law directly addressing the matter as well, but the court's opinion gives a great deal of insight into the history of the question of copyright case law and the requirement of a human author.

In *Urantia Found. v. Kristen Maaherra*, the courts were presented with a question of copyright infringement for the digital redistribution of a work that the parties claimed were written by spiritual entities. Much like the authorship of works by artificial intelligence, this presents the stated issue of creation by a non-human element. However, the court in *Urantia* granted the copyright because it noted the compilation of these writings by people created a unique and original work that was then valid for copyright by the people. Perhaps a similar approach could be taken in certain instances of artificial intelligence creations, such as a compilation written work in which a certain portion of editing, revision, and human created writings were made by a human author after an artificial intelligence generated the initial tale.

A similar discussion to arise from *Perlmutter* is that of the evolving technology being covered by the intent and readings of the U.S. Copyright Act. A specific reference to come from this case was that of *Burrow-Giles Lithographic Co. v. Sarony*. While a particularly old piece of case law, it highlights a heritage to the question of new technologies and copyright. The decision of the court in *Sarony* showed that even though all a camera does is grab a representation of whatever it is pointed at, the photograph is a unique and original work of human creativity and gives the author, not the device or the manufacturer, copyright claim to the works they took pictures of. It also shows another example of the human element receiving copyright for works they claim to have come through another source but that was manipulated enough or represents enough of the human creativity to be warranted protection.

A certain degree of human manipulation beyond original creation is a warranted response to the new paradigm that artificial intelligence gives us in the world of intellectual property. The best material I have found for reviewing how to look at works generated with the usage of artificial intelligence is provided by the U.S. Copyright Office in the form of their denial of four works created with the aid or entirely through the usage of generative artificial intelligence. I have already discussed at length the denial of *A Recent Entrance to Paradise* as it was the central focus of *Thaler V. Perlmutter*. The decision in *Perlmutter* largely reflects what was written in the denial letter. The denial does bring up a point that I would like to discuss a bit further though, in that it further highlights that the U.S. Copyright Office and the Copyright Board lack the persuasive power to be the final word on artificial intelligence and should be bolstered by the efforts of legislators via clarified statutory regulation and definition on the subject matter.

According to the denial, *Thaler* made the claim that the U.S. Copyright Office should be granting copyright to artificial intelligence as this would best serve the public interest, and the Board replied, "The Board must apply the statute enacted by Congress; it cannot second-guess whether a different statutory scheme would better promote the progress of science and useful arts." In this response, they specifically cite the decision of *Eldred v. Ashcroft*. *Eldred* outlines that Congress's authority doesn't interfere with the First Amendment in this issue as copyright protection for your expression is being offered additional protection against being stolen or reproduced but consequently limits your rights when reproducing or repeating the expressions of others. It highlights that Congress is empowered to enact laws regarding those protections, may set appropriate timelines



for the protections, and may create guidelines as to what is given protection or not and in what capacity those protections are enforced.

This reads to me as the Board admitting that they are able to do only what they can within the limits of their statutory authority, and in fact needs permission from legislators in order to change or redetermine their stance on artificial intelligence and copyright. Congress has the ability to determine if policy should reflect artificial intelligence as a tool of the human expression and to what degree of manipulation the tool requires in order to give copyright protections to the works generated using artificial intelligence, and only Congress can change the answer to the question as stands. The U.S. Copyright Office simply lacks the authority to set public policy at the same level as Congress.

Within the currently established bounds, I would at a general look agree with the decision of the U.S. Copyright Office's decision to revoke a portion of and alter the copyright of Zarya of the Dawn. As a general stance, the decision removes protection from any of the artwork inside of the panels that was generated using Midjourney's artificial intelligence and solely grants copyright protections to Zarya of the Dawn's entirely human created portions, which is to say the arrangement of the panels within the comic book and the storyline that was written by Kristina Kashtanova.

However, I find the arguments made by Ms. Kashtanova's legal representation to be compelling as to the nature of human interaction being required to warrant copyright. Kashtanova may have been working through prompts to generate the images, but she underwent significant revisions and reworking of the artificial intelligence's generated images in order to find specifically the image that suited her creative vision for the piece, her representation highlights the interaction with Kashtanova and the artificial intelligence to achieve "Zarya Holding a Postcard". While Kashtanova may have received a four image response to each prompt, she had to clarify and make creative decisions on if she wanted just hands holding the postcard, wanted Zarya's face in the picture, what the background should contain, position within the frame, and a number of similar steps that while they may not have been direct manipulation, does show that Kashtanova had to perform executive functions in the generation of the final image that she then manipulated from there to achieve her final first pane to the comic book. The U.S. Copyright Office dismisses such actions though, claiming instead that the prompts may constitute decisions on the part of Kashtanova, but the way the prompt is broken into tokens that the artificial intelligence then uses in its algorithm represents an element of chaos in the process that the generator cannot predict or control that renders the other decisions that they made along the way invalid.

I would argue though that such an element of unpredictability is valid within currently existing law and would make the comparison in line with a photojournalist or a movie director. In both cases, there is an element of chaos that is almost essential in the creation of their final work. A photojournalist can make the executive decisions as to what events to attend, what location to be at when, angle and timing of their photograph, but when an event is occurring, they lack control over the participation of their actors, the setting around them, and the action in occurrence.

There is a famous photograph named "Falling Man" that was taken by Richard Drew on the morning of September 11, 2001 that I would use in highlight of this example. Drew admits in interview with CBS that he observes and photographs what is occurring around him, and that he will just point and let the camera capture as many images as he can and will sort through them later. Drew has a collection of images taken on the morning of September 11th that feature numerous people in the act of jumping and falling out of the collapsing towers and in various positions during their fall, and the primary reason for the selection of "Falling Man" was that his camera happened to capture the man's decent at a moment of complete vertical upside down from normal orientation. While the curation element is different, in that Drew is curating his selection from photographs of events that occurred in front of him and Kashtanova from a series of images created in response to prompts, the element of unpredictability remains the same. Had any portion of the series of decisions made by any numbers of parties in the events of September 11th been different, then Drew's photograph wouldn't have been taken, and similarly, if Kashtanova had used a different series of prompts, then the base image for "Zarya Holding a Postcard" wouldn't have been generated.

To my other comparison of a movie director, I believe the analogy is more reflective of how the unpredictability is resolved by the slow but significant curation of executive decisions between the initial decision to create a movie and the final outcome. For instance, when casting, the film crew may have some direction and vision for

the role such as: the leading character is male, from 3rd century China, a farmer that was conscripted into service of the Liang army, and should have either a skill in acrobatics or the martial arts in order to be able to perform the scenes in the movie. In this particular instance, the film was produced and written by Jackie Chan, so they knew who the lead was going to be. However, had the role been written without Mr. Chan envisioning himself as the actor, or with another studio, auditions would have been conducted in order to find the best fit for the artist's vision. With those directions, a casting call could have seen the directors picking between Jackie Chan, Jet Li, Donnie Yen, Chia-Hui Liu, or any number of less famous actors that fit the call of male, ethnically Chinese, and able to perform the martial arts and stunts required for the film.

Even during filming, a director can expect and utilize the unpredictability of poor or little instruction to their actors and/or crew. Stanley Kubrick is famous for utilizing this chaos and the choices of his actors by purposefully reshooting scenes ad nauseum and giving vague to little guidance to the actors in order to see what they bring to his vision, perhaps especially for the scripted moment when Jack Nicholson's character in *The Shining* chops through a door, the unscripted element is Nicholson placing his face in the hole left by the axe and yelling "Here's Johnny!".

As decided in *Perlmutter*, works created without the human element lack the ability to be copyrighted, and that would be the separating factor between the examples I just gave and artwork created using artificial intelligence. The unpredictable and chaotic element is other humans and the events of real-world taking shape in ways that couldn't have been decided by the photojournalist or movie director, where Kashtanova's unpredictability is within how the prompt will be utilized by the artificial intelligence to generate the images. In another portion of the U.S. Copyright Office's response to *Zarya of the Dawn*, the Office references *Feist Publications v. Rural Telephone Service* as being the prevailing decision that they cannot grant copyright to works simply because time and effort was spent in their creation if the original material wasn't copyrightable in the first place. I disagree with this reading of the case in part. In the original case, the Court was dealing with a pair of phone companies and the compilation of phone numbers that they made available as a directory for their customers. It does hold up at a glance with the reasoning that currently nothing the artificial intelligence produces is available for copyright, and thus it would follow that anything derived from artificial intelligence would also fail to be valid for copyright.

However, there is a distinction to be made between looking at a compilation of data such as a phonebook, and the artwork generated using artificial intelligence. Justice Sandra Day O'Connor makes note throughout the decision of the Court that the purpose of copyright is to protect the intellectual property rights of the artists and to encourage growth of science, technology, and culture as a whole. She even goes as far as to bring up that a science textbook is a compilation of facts and the facts may not be copyrightable but there are copyright protections for the portions that are not solely facts and for the arrangement of the facts as presented in the book. By the read of encouraging the technology and culture, and that raw data can become copyrightable by the creative expression and manipulation by the artist, I believe the issue requires further clarification by legislators.

Unless the artificial intelligence is generating artwork by simply moving pieces of information from pixel into another to comprise the final generated piece in the manner of a collage, then this issue is on a more complex and almost incomparable scale than what was proposed in *Fiest*. Artificial intelligence does take data points from its programming in order to generate its final work, but these data points are not as simple as a phone number. They are the results of complex algorithms that are themselves the compilation of the training materials, the programming directions given by its creator, and finally the prompt tokens generated by the end user.

Jason Allen's "*Théâtre D'opéra Spatial*" was rejected due to public knowledge that the work was generated using artificial intelligence but Allen had failed to disclose the artificial intelligence use in his application. The letter contains the same precursory language describing artificial intelligence being ineligible for copyright and the requirements for human authorship that are seen in the other letters the U.S. Copyright Office offered in rejection of works containing elements generated with artificial intelligence and which all is baked into the *Perlmutter* decision.

Taking a look at the provided images of what was generated using Midjourney's artificial intelligence and the final "*Théâtre D'opéra Spatial*", it is clear that substantial changes were made to the image, but Mr. Allen

refused to offer explanation of the process and steps in between the two images when the Office inquired further. Without that explanation, I find myself in agreement with the board, but would like to take a moment to address how their rejections of several points in Mr. Allen's argument relate to the discussions of this paper. Firstly, while the comparison can be drawn that the control of the prompts and the control of a garden do both rely on elements outside of the artists control, the Court in *Kelley v. Chicago Park District* takes time to lay out that the nature of plants is to grow, bloom, and decay, and discusses at length these elements that would occur after the art was made are the portion that is outside of the control of the artist and that because those elements mean the art cannot reasonably be guaranteed to remain intact, meaning in the form the artist had designed, that protection would not be valid. The Court in *Kelley* does also note the human requirement in copyright, but as the U.S. Copyright Office establishes this earlier in the letter, I am assuming the focus of this reference was focused upon the elements beyond an artist's control. The Board's rejection of Allen's claim that denying works with artificial intelligence copyright protections leaves a hole that discourages end users from investing in the usage of the tool is in alignment with their previous decisions. As discussed earlier though, this may not be the correct method to handle the issue in order to best encourage growth within the industry, but it is not in the hands of the U.S. Copyright Office to make the necessary change to the rules as it is beyond their delegated authority.

Finally, I do agree with the stance of the U.S. Copyright Office that material generated by artificial intelligence should be disclosed and that this request is not burdensome. Artificial intelligence is not like any tool that artists have ever had for the creation of copyrightable works. Despite my disagreement as to the nature of human manipulation both in terms of prompts and in terms of what is done to the image after the artificial intelligence generates it, it does allow for the bypassing of time, skill, and energy that a visual artist has to take time to develop. Noticing commonalities between Allen's denial and Kashtanova's denial, I would pose that the U.S. Copyright Office seems to be willing to dismiss any amount of work through the prompts and tokens at this time, and may be looking for an element of transformation between the images generated by artificial intelligence and the final product under human manipulation that may be unattainable after the decision of *Andy Warhol Foundation for the Visual Arts v. Goldsmith*.

There is one more denial from the U.S. Copyright Office's Copyright Review Board that needs touched upon before I go into depth about the Warhol decision. A work by Ankit Sahni entitled "SURYAST", a blended image of a sunset that Sahni took a picture of and Vincent van Gogh's *The Starry Night*, was rejected as, in the Board's description, Sahni merely choose two images, one to be the base and one to be applied as an overlaid filter, and then chose the strength of the filter while the artificial intelligence performed the work of combining his selected elements. He could always apply for copyright of the picture of the sunset that he took. The U.S. Copyright Office allows the use of Adobe, Photoshop, and other image altering software. I don't see a substantive difference between what Mr. Sahni did with artificial intelligence and the usage of photo editing software beyond his application of a work that has entered the public domain. Either way, I believe that it fails to be counted as transformative by the Warhol decision.

The matter of Warhol sits almost perfectly as a bridge between my two major questions. However, I find myself in agreeance with Justice Kagan in her dissent that the bar has been placed almost arbitrarily and shows a deafness from the remainder of the Court as to the nature of creativity and free expression. The majority decided the most important question in the determination of fair use was to the intent of the work after creation, i.e. sell for profit, displayed in private or public, or provide commentary on society.

That answer to the question leaves me to pose a thought. The reading of the case seems to imply that *Orange Prince* was made as a commissioned work for *Vanity Fair*, thus made with the sole intent of commercial use. Instead, let's say Warhol had made *Orange Prince* for his private collection, it remained in storage for the intervening 32 years until Prince's death, and then appeared on the cover of "Condé Nast". Would the alteration of the original usage have made a difference in this case? I believe Kagan is on the correct path in her assessment that mere commercial usage needs further digging and explanation before a judgement of the work can be properly determined. If this stance had been taken within *Campbell*, then the outcome would have found 2 Live Crew in violation of copyright as both songs from the case occupied a commercial space in the music industry. After all, anyone could remix the instrumentals, add a beat, and rewrite the lyrics in the same manor that the Majority dismissed Warhol's work as being so simply redone.

Kagan does an excellent job of summarizing that mere commercial interest is no longer a complete picture of value or expression in assessing *objet d'art*. In the instance of Warhol, the further question should have been asked as to the expressiveness of the works and the appropriate marketability of the works in competition with one another. Orange Prince gives the impression of the largess of Prince's persona, continues coloration and stylistic decisions similar Warhol's other Orange works, and uses the lining and minimalism to reflect on fame and shift perspective on the subject of the work. While Goldsmith's photograph utilizes black and white to maintain seriousness, head and eyes of the subject angled to imply thoughtfulness, and overall, the work focuses on Prince as a person rather than an icon. Much in the same way that "Pretty Woman" and "Oh, Pretty Woman" convey different messages and occupy different regions within the commercial space, I believe Warhol and Goldsmith's works attract similar markets in the focus of Prince but convey different impressions and will naturally draw different sectors within the marketplace.

I would join both the majority and the dissent in referring to the closing thoughts of Justice Oliver Wendell Holmes Jr. on the matter of *Bleistein v. Donaldson*. Justice Holmes warned that those who focus on the study of the law are ill suited to assign value and importance to artwork. They will miss important pieces, elevate works in accordance with their tastes, and hamper the open and free development of culture and creativity. I find this relevant to both portions of my discussion. Judges and the U.S. Copyright Office would be beyond their assigned powers to grant copyright to works generated by artificial intelligence as they have no statute to base it upon, and are operating out of their depth in making decisions as to a precise amount of work required in order to elevate a generated image into something that a human could effectively claim copyright upon.

The second big question to arise from this paper is what the copyrights of original authors are when their work is used in the training of artificial intelligence. There appears to be a distinction to be made in this question that warrants a bit of discussion at first. The method of integrating art into the training of artificial intelligence does matter in making a determination. Is the image stored and referenced within the programming of the artificial intelligence, or is an algorithmic representation generated that is then merely a portion of the coding and not identifiable as the original copyrighted work? In a form of layman's, does the artificial intelligence have a picture that it looks at to borrow from, or is the artificial intelligence taught the techniques and styles of the picture that it then draws on from memory?

This distinguishment could be an easy method for determining a comparison to human creativity and thus determining if the training methods for the artificial intelligence is infringing upon copyrights or if they are more akin to the fair use methodology for teaching an art class. Such distinction placed in law would grant a *prima facie* defense to artificial intelligence companies and end users that any infringement would exist in the database the artificial intelligence references for the creation of new works rather than the artificial intelligence itself. In inverse, it would also then require the artificial intelligence programming to be available for review in an accusation of infringement. I make note here that it would need to be a new law or a new understanding of the paradigm as teaching a human and teaching an artificial intelligence may be debatably similar, but it is not an equivocal balancing. Firstly, an artificial intelligence is programmed or trained using an algorithmic model that while one may call it learning functions in a very different way than the human brain during the act of learning. Secondly, the purpose of fair use is expressly to promote the advancement of science and culture by not restricting the market, and while the exception would advance science by allowing further development of artificial intelligence, it could also stifle and restrict creative works from humans that actually need the protection in order to benefit from their intellectual labors.

Artificial intelligence that merely references an external database does grant an easier determination of fault that would fall in alignment with the precedence set up by *Viacom International, Inc. v. YouTube, Inc.* and further clarified by *Obodai v. Demand Media, Inc.*. These cases asked if a public media website that allows users to upload content are responsible for copyright infringement present on their domains. The determination was that such websites must take reasonable action on their own and investigate/comply with cease-and-desist orders in order to be clear of infringement claims. If there is reasonable evidence that they either failed in such actions or had and hid knowledge of infringing content being uploaded, then they would be liable for the infringement in the same way as the person that had uploaded the content to their site. *Viacom International, Inc. v. YouTube, Inc.* had the primary failing of human error. It wasn't their monitoring policies or guidelines that ultimately gave YouTube liability against copyright claims. It was the employees that were careless about waiting a day or

two to take down a video, or knowing that a video was a violation of a major media outlet's copyright and ignoring the matter. The major difference for *Obodai v. Demand Media, Inc.* was a clearly laid out and established policy for addressing copyright claims including informing all users that uploaded to the site that the monitoring occurs and can result in termination for even a single complaint of copyright infringement.

Straying momentarily from legal discussion to touch on the technology, if such an algorithm has not already been developed and implemented on a social media platform, I believe it would be reasonable to set up a system in which an artificial intelligence could handle claims of violation, monitor removal and usage, and report outcomes to the developer for review. It would be an unbiased monitoring algorithm that would be able to be more effective and efficient in keeping the database clear of violations because it can be programmed with precise guidelines and can learn to more accurately address the issue as time goes on.

The road to determination if the work is located internally and not as a copyrighted replication of the art becomes much more twisting and unclear. In reviewing the various filings and decisions attached to *Anderson v. Stability*, *Tremblay v. OpenAI*, *Doe v. Github Inc.*, and *Guild v. Open AI*, it quickly becomes clear that the old understanding of looking for direct copying either in form and essential elements or for an actual copy of the work is not going to be workable going forwards with this artificial intelligence and the training models being designed for generating artworks.

An important side note to anyone that may be considering attempting to file an infringement claim against anyone for any reason, you must register with the U.S. Copyright Office and receive approval of the registration before you can launch of suit for infringement. Also, always read the terms of service fully for any software or service you use in attachment to your art, in *Anderson v. Stability* a portion of the claim that involves believing that DeviantArt was in breach of their own terms by implementing an artificial intelligence program into its website was in fact covered by a notice in the terms of service that cautioned by agreeing to the terms that third parties have access and ability to copy any work displayed on the website and that DeviantArt can reproduce and create derivative works of anything uploaded to the website. Further, DeviantArt claims, and Judge Orrick agreed, that despite implementation of an artificial intelligence on their platform, they are not directly responsible for its training and are merely providing a database service which would have moved them back to a YouTube or Obodai scenario had the claim not been dismissed.

A second decision by the District Court for the Northern District of California was reached on August 12, 2024, which provides further clarity on several of the issues that should be considered by lawmakers for the inputs of artificial intelligence. Firstly, there was a claim towards induced infringement based on a statement from Stability's CEO in which the belief was propagated that because the engine was trained on various artists works that one could recreate the training materials through the use of prompts within the system. If proof can be established that such reproductions could be manufactured using the artificial intelligence, then the argument for promoting infringement by their users would be substantiated and could require companies designing artificial intelligence in the future to establish safeguards in future models that would prevent them from reproducing, either directly or substantively, any of the materials used in its training.

Secondly, Judge Orrick is allowing the training question to continue forwards. If there is no evidence that a direct copyrighted version of the works is in the coding of the artificial intelligence, or if Judge Orrick will allow an algorithmic representation of styles to be counted towards a copyright claim remains to be seen. Both have major implications upon the future development of artificial intelligence and intellectual property law. While it may not matter what format of distribution is used to consider a movie a protected movie, I believe that the essential substantive elements must still be identifiable for a claim to be valid. For instance, let us say that I study *Enter the Dragon* and *Drunken Master* in order to understand martial arts cinematography, and then write and direct *Fearless*. Despite all three films being martial arts films, and having some overlap in technique for choreography of the fight scenes (open palm strikes, various throws and locks), there would be little else in common between the works I studied, my inputs, and the film I produced at the end.

Judge Martinez-Olguin dismissed portions of claims in *Tremblay v. OpenAI* that could shape or be shaped by a decision after determination of the distinction of algorithms being covered or not. She dismissed claims to intentional removal or alteration of copyright management information, vicarious copyright infringement, and distribution of works or copies all for the reason that substantive evidence was not shown that the artificial intelligence could effectively reproduce any of the copyrighted works that were claimed to be used in its

training or that the books being converted into algorithmic information was the same as taking a digital copy of the works in question.

Between Tremblay and Andersen, there will be a precedence by the end of their cases and appeals as to the burden of evidence required to show similarity with copyrighted works from the outputs of artificial intelligence in claims of infringement.

Throughout Andersen, Judge Orrick shows an appreciation for the need to take a new look at all laws and decisions that could be relevant to the use of copyrighted works or of infringement upon said works in the process of training artificial intelligence. I agree with this mindset due to the unique nature of artificial intelligence as a tool for the creation of artistic works and the way it is substantially different than any previous method of production in this field. Similarly, there is a sense in Andersen, Tremblay, and Guild that the parties claiming to have been victims of infringement are less concerned with any individual piece of their work being reproduced than they are about the ability of artificial intelligence to reproduce their individual styles. It seems at its heart be a concern for a theft of the quintessence of their uniqueness.

One possible venue for exploration could be the U.S. Patent Act for some artists, provided that an artist's unique method for the creation of their work can pass the Mayo /Alice test as discussed in the U.S. Patent Office's guidance on patent subject matter eligibility. It would need to be more than just the idea and form of their work though as that may not be substantive enough to rise to the level of patentability. This would also rely upon a decision from Andersen or a related case in the future as to whether or not an algorithmic representation of style is substantively the same as a digital representation for the purpose of an intellectual property claim of infringement. However, an argument could be raised that while persuasive, a decision from Andersen could be seen as not binding to other forms of intellectual property as they are in fact separate sections of law, which again calls for legislators to address the issue in a more binding manner.

There are those who simply won't rise to the standard of a patent claim and it is possible that a copyright claim may also be found to be invalid dependent upon how the courts decide on the matter of format and the training of artificial intelligence.

In looking at the concern of the parties in the previously mentioned set of cases about the theft of the essence of their uniqueness, I would encourage lawmakers to look to the efforts being made in addressing digital replicas and similar name-image-likeness laws as they are focused similarly on the overarching concept of having what is unique and defining to an individual and offering protection against the theft and other forms of harm posed by artificial intelligence and replication of those qualities. The U.S. Copyright Office issued an in-depth report making recommendations on artificial intelligence and copyright focusing on digital replicas in July of 2024. Representative Junie Joseph of Colorado, and Senators John Cavanaugh and Eliot Bostar of Nebraska have made efforts at the state level to address the rising concern about digital replication and Senator Christopher Coons of Delaware and Representative Maria Elvira Salazar of Florida have introduced the No Fakes Act of 2024 in the House and Senate respectively.

Laws such as the No Fakes Act are primarily focused on reproduction of a person's image and vocal patterns in the effort to stem the creation of "deepfakes". The heart of both the matter of deepfakes and claims of copyright infringement during the training of artificial intelligence are the same, having one's personage stolen or misrepresented without any means of recourse. The U.S. Copyright Office's Report takes time to highlight that there are a number of faults in current efforts from being too narrowly tailored and failing to fit every venue of intellectual property law to pre-emption issues due to difference in various state and federal laws.

Within this paper, I have taken a dive through the current stance being taken by the courts and the historic beliefs on the issue of copyright and artificial intelligence. I have shown that our species has been the sole creator of imaginative works and thus the sole recipient worthy of protection to their rights as creators. I have found that the method of programming can be seen as a division between that protection and free use. The case law and pre-existing statutes upon the rights of human creators has given them a very wide array of claims to protection and usage. However, without clear legislation that establishes in plain language the rights of creators and the rights for usage of generated works, the risk to the public of either having the fruits of the intellectual labor stolen or of being sold rights that simply don't exist.

I am proposing some guidelines for the creation of new legislation which would fill the void of having not yet declared that our species as sole beneficiary and why. I would recommend following the historic stance of

humans are the only ones who require the protection of copyrights. Firstly, it would need to clearly state that no matter what methodology is used for training the artificial intelligence, all human created art has the benefit of copyright for the training of artificial intelligence and should outline the pathway for artists to properly exercise that claim to address the work being used in the training of artificial intelligence models.. Works that have entered public domain or those generated by another artificial intelligence would be of fair use for companies to use, but otherwise they must acquire rights or permissions as appropriate to each individual piece of art. Secondly, a final and appropriate determination in following with previous alterations allowed without violation of copyright to create a new calculation for a requirement of human alteration in digitally generated works to allow for a protective space to be determined for those who would use this new medium to generate art. There is a no hard and fast rule as to the amount of alteration needed to elevate a work to be used outside of the original owner's copyright. Instead, determination is made on a case-by-case basis to find whether infringement was intentional, meaningful, and substantive. Creating a hard rule on amount of change needed for an artificial intelligence to go from unprotected and machine generated to being a tool of assistance in creating copyrightable works would create motivation for industry stakeholders to continue and expand investment. As I have clarified previously in reviewing caselaw on the matter, creative industries and their representatives want to have artificial intelligence as a tool in their markets and are open to a dialogue about how to make it viable and functional.

A number of times during my research, I came across a quote used from a review of the American Law Institute's 2020 draft for a Restatement of the Law, Copyright by Justin Hughes. The quote is used dismissively of artificial intelligence and the discussion of copyrights for works generated using artificial intelligence, remarking that, "The day sentient refugees from some intergalactic war arrive on Earth and are granted asylum in Iceland, copyright law will be the least of our problems." This statement sounds like a wave of the hand, and the quote is used as if to say to the reader, "It is a silly flight of fancy to think about law for science fiction." However, Mr. Hughes continues from his statement on intergalactic refugees to point out that after nationalization was settled, certainly the aliens would be folded into our copyright laws. I believe Mr. Hughes' intent in this statement was in alignment with the remainder of the statement. It is simply unnecessary to fully address the rights of any sentient being besides humans in the matter of intellectual property because humanity hasn't encountered a sentient being that wasn't one of us. These conversations are important as the technology continues to progress. In examining the difference in need and usage of protections to their intellectual labors between humans and artificial intelligence, we also are forced to discuss what the protection means and establish guidelines for determining whom those worthies of protection are. As the computing power and ability to learn moves artificial intelligence forwards towards actual thought and imagination, we will have an understanding already established and can be adapted to an appropriate framework for a synthetic intelligence's rights.

A final thought I would direct towards the developers of future artificial intelligence and those who handle the training of those next generations. While the legal question towards the usage of copyrighted works may indeed prove to be allowed due to the law and the technology being out of synch with one another, there is a moral question to ask as to if the usage of these works should really be allowed. I would instead encourage those that train artificial intelligence to perhaps take a leaf from the music industry and treat the works in your training as though you were sampling music for your next track. There is an analogy to be drawn as both acts use portions of a larger copyrighted work to make something new for the industry. Compensation can be different than using the whole work, but you will have then shown respect to the efforts and process that the artists and creatives whose work you are borrowing from deserve.